TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.0 ± 3.69
Correlation Stability	82.7 ± 0.78
Distribution Consistency	86.5 ± 2.97
Entropy Preservation	52.1 ± 0.83
Mean Z Score Drift	0.3 ± 0.05
Categorical Fidelity	90.5 ± 1.96
Privacy Risk	0.1 ± 0.30
Data Density	100.0 ± 0.00
Coverage Score	93.2 ± 2.23
Feature Accuracy	74.4 ± 5.73
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	30.378	-14.368	32.1%	Drift
TB	3.299	2.382	-0.917	27.8%	Drift
DB	1.486	1.456	-0.030	2.0%	OK
Alkphos	290.576	365.288	+74.711	25.7%	Drift
Sgpt	80.714	85.260	+4.546	5.6%	OK
Sgot	109.911	29.693	-80.218	73.0%	Drift
TP	6.483	7.062	+0.579	8.9%	OK
ALB	3.142	3.603	+0.461	14.7%	Warn
A/G Ratio	0.947	1.101	+0.154	16.3%	Warn
Selector	1.286	1.358	+0.072	5.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.0 ± 3.69
Correlation Stability (*2)	82.7 ± 0.78
Distribution Consistency (*3)	86.5 ± 2.97
Entropy Preservation (*4)	52.1 ± 0.83
Mean Z Score Drift (*5)	0.3 ± 0.05
Categorical Fidelity (*6)	90.5 ± 1.96
Privacy Risk (*7)	0.1 ± 0.30
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	93.2 ± 2.23

Feature Accuracy (*10)	74.4 ± 5.73
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6111111111111111	0.7042735042735042	0.02991452991452991	
F1	0.7821343746683586	0.7378676814861126	0.8131532913885682	0.03101891672020953	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7145299145299145	0.6384615384615384	0.7119658119658119	-0.00256410256410256	
F1	0.8154972140649285	0.7696699450886451	0.8223906038363822	0.00689338977145373	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6854700854700855	0.5931623931623931	0.7094017094017094	0.02393162393162393	*
F1	0.7909408486522184	0.7246793711902915	0.8103933744468927	0.01945252579467417	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6897435897435897	0.5897435897435898	0.6538461538461539	-0.03589743589743588	
F1	0.8011016273174306	0.7181112640332852	0.7626494363356547	-0.03845219098177584	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6529914529914531	0.7128205128205127	0.00683760683760846	
F1	0.8128002697439015	0.7824287547551977	0.82887063646441	0.01607036672050851	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7	0.7068376068376068	0	
F1	0.8279091624529537	0.8226693841703469	0.8279091624529537	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5700854700854701	0.6854700854700855	-0.01709401709401709	
F1	0.7997903571549948	0.6996844130006588	0.7850045084784452	-0.014785848676549607	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743595	0.5675213675213675	0.4982905982905983	-0.06068376068376067	*
F1	0.5587432119380917	0.6504823518347967	0.4600246733120735	-0.09871853862601787	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6641025641025642	0.5829059829059828	0.6752136752136753	0.011111111111111094	
F1	0.7663032365208943	0.6980454655103909	0.7788302012336018	0.012526964712707523	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6435897435897437	0.6743589743589744	-0.011965811965811979	
F1	0.7971299319768493	0.7593460076233256	0.7894737426800507	-0.007656189296798688	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	-1.247037149307881	24.413777037494093	313	350	-48	6.807889020241033	3.956493334512949	1.1599536217538389
2	29	Female	1.6042531311015775	0.3362893558038697	260	83	29	8.402352111889952	1.914037669305869	1.342519335047852
3	37	Female	-0.20793816681531418	0.3651383282215753	104	94	87	7.581431866300245	2.264469885347714	1.289228497647131
4	31	Female	1.6819600554193723	0.8434646907030241	525	172	-30	7.124573323735049	3.832207014425768	1.615648153139947
5	10	Male	6.852609745198022	0.4036951602902953	101	100	184	5.993033545210973	2.469993566330122	1.42312916626007
6	56	Male	0.05651151184265091	4.241147415757298	326	74	-49	6.608556866844355	4.156996172922243	0.8333680379318431
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	25	Male	0.8660034926900039	0.9671591923824017	110	-3	1084	6.330940391727962	2.337183024964479	0.7237386477000026
2	13	Male	33.329110674228716	17.989530405368864	259	48	133	4.832204243412629	1.103196352548046	0.6123482815630142
3	60	Female	14.677606824012667	20.357719638270627	172	28	100	1.741058776625614	3.801365519754098	-0.015388145409669818
4	49	Male	0.6302848812996347	-4.017352838612661	297	77	51	5.580482904779141	3.140649086634601	0.7156287898569497
5	62	Female	-0.18180805114010434	0.3905583708763327	376	227	544	3.735563372011861	1.271743175266379	0.5926460524097809
6	12	Male	1.526020009981557	2.556218885301549	326	303	139	4.33654417607331	1.630642621995742	1.1391894903070445
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	8	Female	-3.0714782505623885	2.0802686860941697	418	47	127	6.900208232565635	4.586739232613495	0.9540161349500917
2	38	Male	0.6075440326430949	1.9070047624649846	217	99	95	4.986894101076453	4.143819312245923	0.6856898084190658
3	57	Female	0.9495907910392262	1.7545364235656975	232	-22	95	9.14196807119785	2.952526029619386	0.5004732464562316
4	8	Male	0.007014926761611973	-0.7426620450469257	143	13	-15	6.66328454030603	4.663242730420125	1.0356913696856112
5	49	Female	1.1851579944979753	0.4884766383502847	95	-39	103	7.618971347035756	2.219757686229304	1.0488498236204389
6	65	Male	0.7925429564070519	1.6526476392390412	-58	-33	42	7.109543134961311	2.583348886861708	1.0431960702784209
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Male	1.5481677776830443	-0.16589657949127335	175	9	162	4.367514074784051	3.16082749669823	0.1989517849242932
2	43	Male	0.2250323898470315	1.6631731372073983	879	50	85	6.089970455039927	4.314042719230727	0.4582786313843763
3	65	Male	2.629158682887027	0.7467050624403007	183	1758	40	4.670020020845341	3.306131300109343	0.8839656522868132
4	38	Female	2.25748759568269	0.1903707904693822	166	10	-33	5.393042813083534	3.268653226793751	0.6789344668581912
5	54	Male	0.4803839430800177	1.3313832876771605	338	53	26	4.608961649859898	4.010976985923083	0.6527152200472119
6	62	Female	1.1242148885750278	-0.23693988985590148	263	31	46	5.523063148933131	2.997135053372241	0.7090199691406297
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	16	Male	23.4636035200094	-0.02314129734830317	59	60	4	6.787619789038653	3.56533514043137	0.930451899753393
2	7	Female	1.9574932901747024	2.302820686426258	202	18	-11	6.035007891538322	3.343264002669847	0.8726801742504373
3	60	Male	3.6897903907483833	1.2305302334009118	518	85	64	5.689792504987396	3.728793892297547	0.12219466175251714
4	41	Female	1.753050079279355	16.888673138266974	166	16	-27	4.81102576297061	2.3293832913406747	0.3907346362190387
5	20	Male	0.6696000042148085	4.461521687679906	264	249	4	5.57104654029244 2	4.277916744957669	0.2776839417017386
6	33	Female	0.3736565205479099	0.8009310732186516	182	997	152	5.412972350005787	3.4046852493354227	0.5354100830495253
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	15	Male	-0.18729726121710244	0.15561108543914165	285	46	17	7.28905232338205 1	3.67383233697057 45	0.5642499946411 37
2	59	Male	5.996427066265108	0.35994883263306 2	201	34	37	6.52379414956456 4	2.48616092609229 83	1.2789913127039902
3	31	Male	0.6165733252252391	0.6408525544117737	429	19	15	2.63904210102787 26	1.53597096020332 91	0.5336796537467 05
4	18	Female	2.0158786923220253	-0.38915352735854597	276	53	63	8.08453518858419	4.27146176203643 6	0.5403186100295893
5	10	Female	30.611597817802142	0.9834840912752754	237	47	-44	6.22379220384528 5	2.34906549138461 25	0.4392741010332 2534
6	25	Male	-0.7630172032284055	0.10509774951479872	1131	66	141	1.37099626215602 03	2.75416826964841 07	1.0273327470542852
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	2.161020148808852	-0.5612548851316618	269	75	33	7.247590062514033	2.514511828715812	0.7555751437576116
2	64	Male	1.5110746621799322	-0.21442986146563314	276	615	139	8.22638236031013	2.2776525510797985	0.9267344840365407
3	66	Male	1.6233680955481855	-0.47214252556754854	233	15	50	8.26669083963052	3.372500989642921	1.2520665438431382
4	-3	Male	2.595216813362314	0.011474135020665965	232	45	178	6.12582771937288	3.2806937459128873	0.1648584753686467 3
5	82	Male	2.334229080720365	-0.33498856958132395	184	-16	80	6.09953021356967 5	3.2854181245396723	0.3755677300232107 6
6	31	Male	0.07470970338384386	0.23282988845332472	300	253	347	8.096520241808456	3.242240617351645	0.7579161028433891
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	41	Male	7.274250434655621	0.061954751220659854	264	-2	68	5.222575096754396 5	3.89572270112463	0.7396048066765191
2	56	Male	1.0130715494947307	0.07529141857065724	240	61	96	4.689811667218202	5.15814627805956	0.8664955911422075
3	35	Male	-0.514358200752147	-0.44699465190643334	210	8	328	5.651642989401139	3.232616165688571 5	0.4483083305374422
4	4	Male	-0.45328357587642 6	0.19190185232128698	91	73	42	7.909251543939674	4.886449768776851	0.7656035243063438
5	32	Female	1.1780048713590383	-0.437530206795764	101	48	58	7.479259666023116 5	2.882260757696447	0.9457318984024281
6	30	Female	1.0491695201287907	0.24805789952547216	193	0	-14	5.342087831762579	2.819722884311169 2	0.7055681277668513
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	47	Male	0.964231507518974	0.658942857248348	134	29	273	5.823937955365721	2.656478972022816	0.425743836537243
2	8	Male	2.486810391990507	1.156309212110669	564	-12	370	7.415180773113458	2.257961419747561	0.627602508480618
3	35	Female	2.366139630261254	0.370564621899029 94	273	17	91	5.722449504387984	0.689281889576136 6	1.054414388063474 4
4	15	Female	5.448648271933731	0.746879024815979 5	126	30	85	7.148869185818968	1.961121735797665	0.961036955585513
5	2	Male	3.037300065328444	0.617496894754106 6	440	16	57	6.341063315356486	2.631368282609434 7	1.292841264928425 3
6	24	Male	6.074832585903485 5	0.286578034807076 3	234	39	35	6.665113892261369 5	3.245381591625431	0.743932943449400 2
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	64	Male	6.271512711403385 5	0.311092471689417 1	197	-4	68	4.230667499076746 6	3.998170908309523	1.105829178208807 9
2	73	Female	0.173498316680965 6	0.029337888692469 283	208	-4	114	5.012558218688373 5	4.006028468222563	0.681526757526090 8
3	62	Male	2.897179798865994	3.645240201308172	152	34	100	4.163157527796271	3.878445976191752 5	1.117242191938570 4
4	58	Female	1.150157186995162 3	0.198168597688691	204	217	53	3.399430354320826 6	3.185014317933806	0.776208928411376 9
5	31	Female	3.182936700331996	0.529393749567565 4	538	7	90	6.546331628685678	1.407815908006234 2	1.813225093966714 7
6	64	Female	1.405508804467890 3	0.667247593606856	270	2	68	3.200595115895019	1.621229934745261 7	1.783811879835859
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3